



Communication/s Networks
Research Group



Advanced
Grant
GO-SPACE

Taming the Heavy Tail: Age-Optimal Preemption

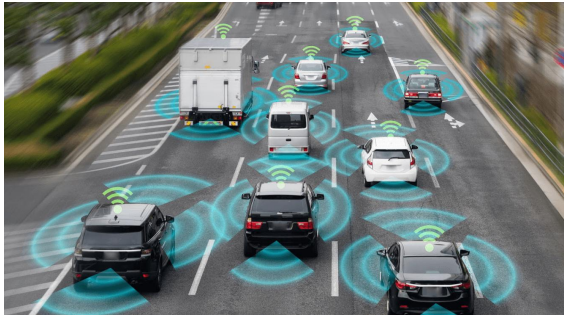
Communications Networks Research Group (CNG)
Middle East Technical University

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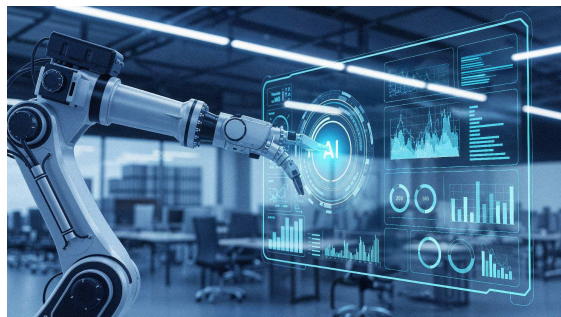
IEEE ISIT 2026

2026/6/30

Freshness is Not Delay: What is Age?



V2X Communications



IIoT Applications

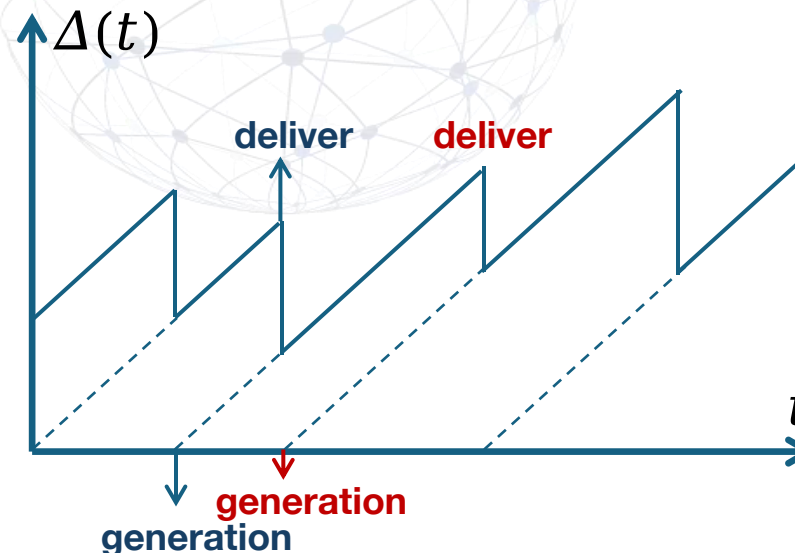


Digital Twins

Age of Information (AoI)

$$\Delta(t) = t - U(t)$$

- $U(t)$: generation time of the most recently received packet
- AoI measures how fresh the data is from the receiver's side



Age is a **receiver-centric** metric.

- **Throughput:**
How much data is delivered?

- **Delay:**
How long the packet is waiting to be delivered?

- **Freshness:**
How stale the **receiver's knowledge** is?

What Makes Fresh Information Get Stuck?

Heavy-tailed service delays arise when **sensing, computing, communication, or connectivity pipelines** occasionally stall.

Heavy-tailed service delays

Communications

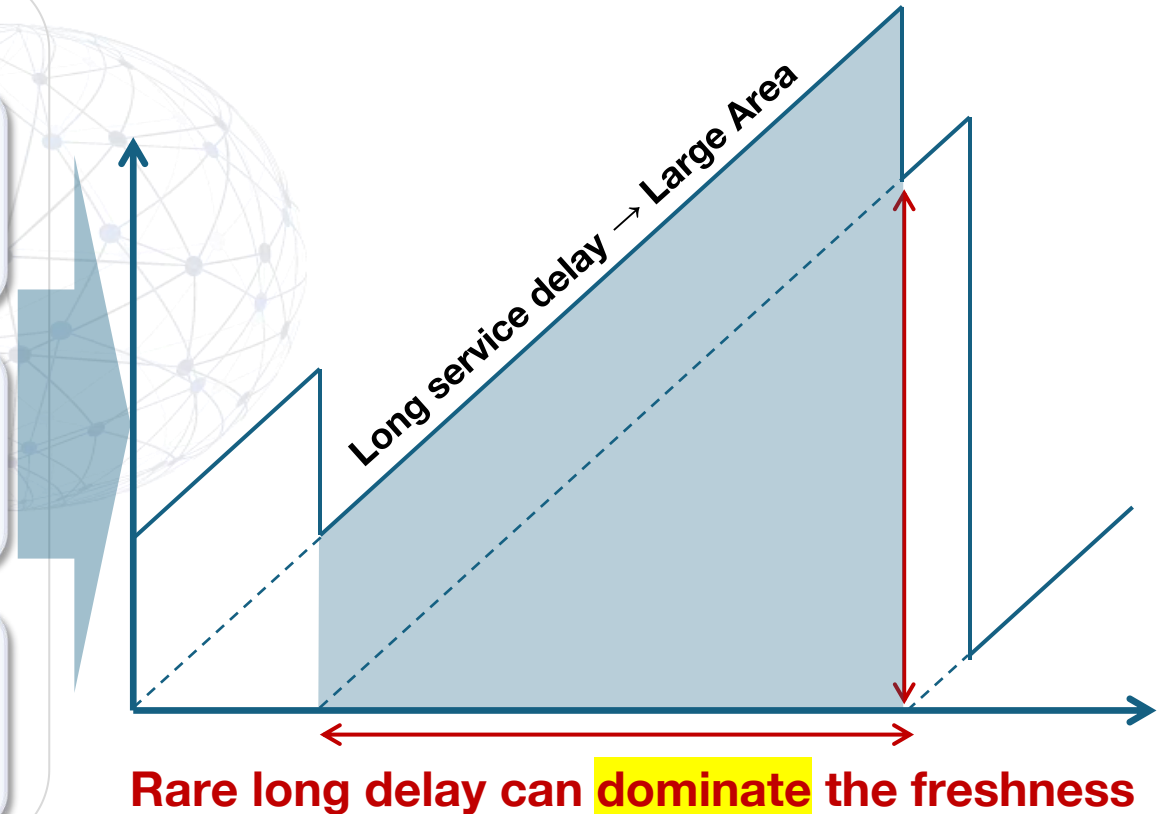
- Queues, routing changes, retransmissions
→ **rare long delay**

Computations

- Edge inference, cloud services, stragglers
→ **variable processing delay**

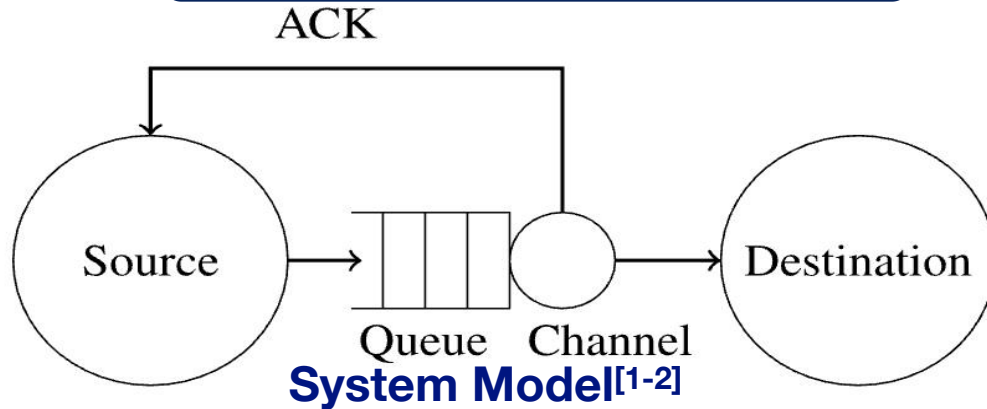
Connectivity

- Drone, space, industrial IoT, intermittent links
→ **long contact gaps**



How to Minimize Age of Information under Random Delay?

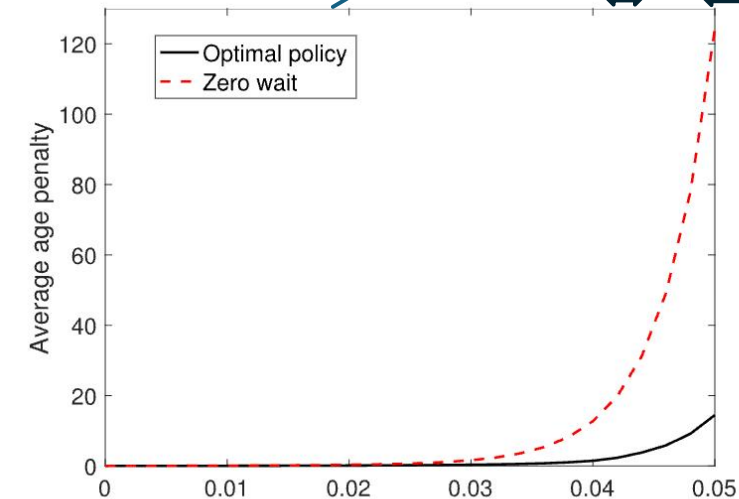
2016 INFOCOMM/
2017 TIT: Update or Wait?



- **Question:** When will the source generate the next update?
- **Surprise:** Throughput-optimal zero waiting is not optimal
- **Principle:** sometimes waiting is better — “lazy is timely”.

The intuition
Deliver and sample

The reality
Deliver and wait



Zero-Wait vs. Aol-Optimal Sampling

[1] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, *IEEE Transactions on Information Theory*, 2017.

[2] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, *IEEE INFOCOMM*, 2016. (**IEEE/ACM INFOCOM2026 Test of Time Paper Award**)

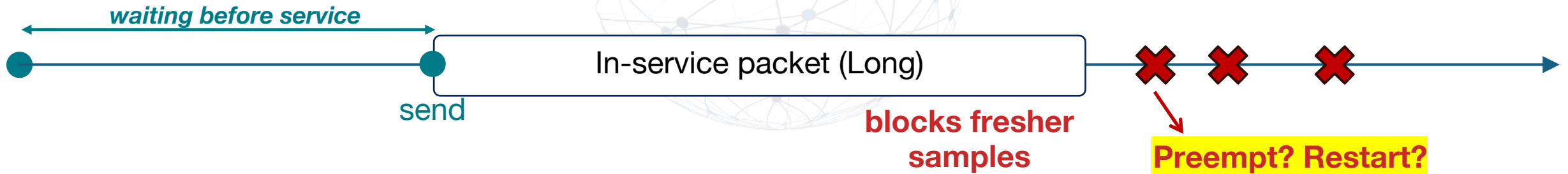
From “Lazy is Timely” to the Preemption Gap

Known Result: Non-Preemptive Result

- Waiting before service can reduce average Age of Information [1-2].
- This yields the principle: *lazy is timely* [1-2].

What is Missing

- Once a packet is in service, the source *waits passively* for ACK.
- Fresh updates are blocked by the in-service packet.



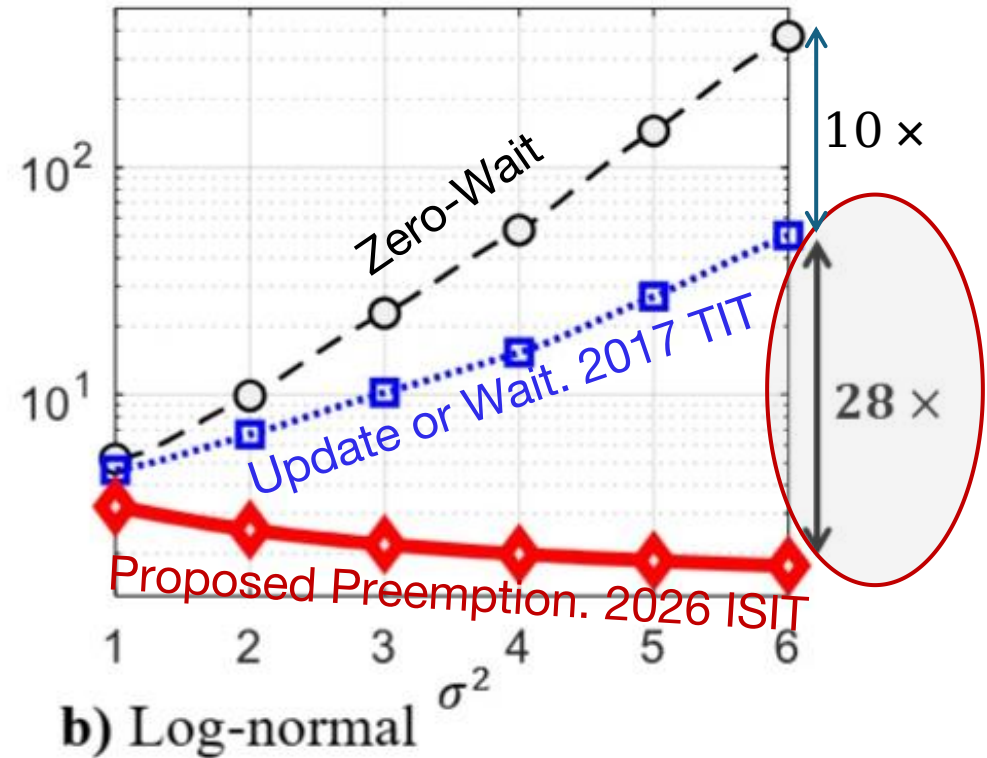
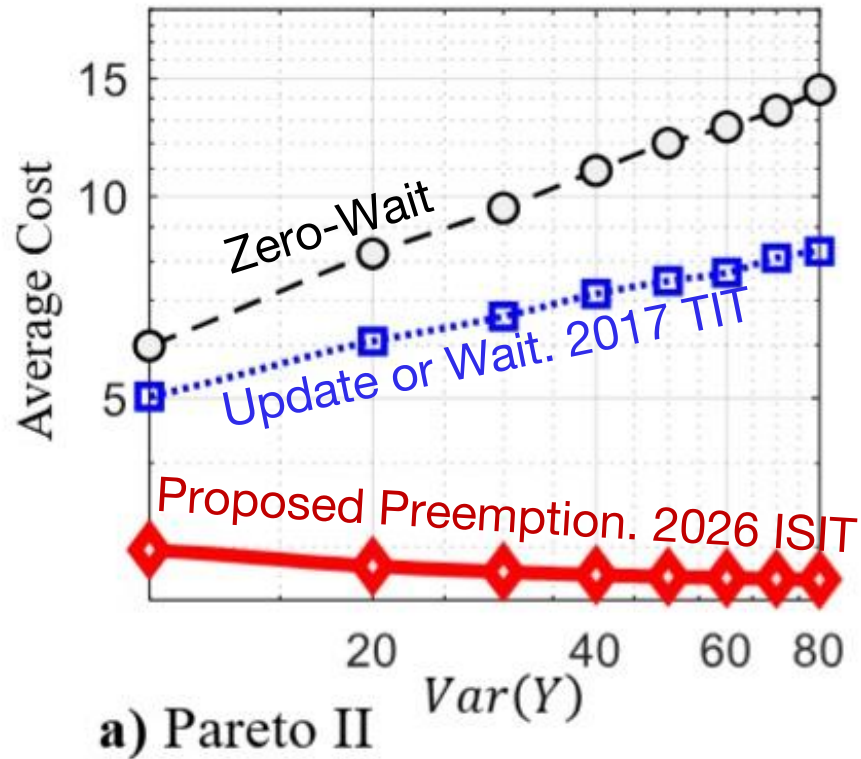
Waiting helps before service starts; Preemption matters after service starts.

[1] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, **IEEE Transactions on Information Theory**, 2017.

[2] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, **IEEE INFOCOMM**, 2016. (**IEEE/ACM INFOCOM2026 Test of Time Paper Award**)

Our Results at a Glance: The Power of Preemption

Our Idea: With age-optimal preemption, high age gains can be achieved.

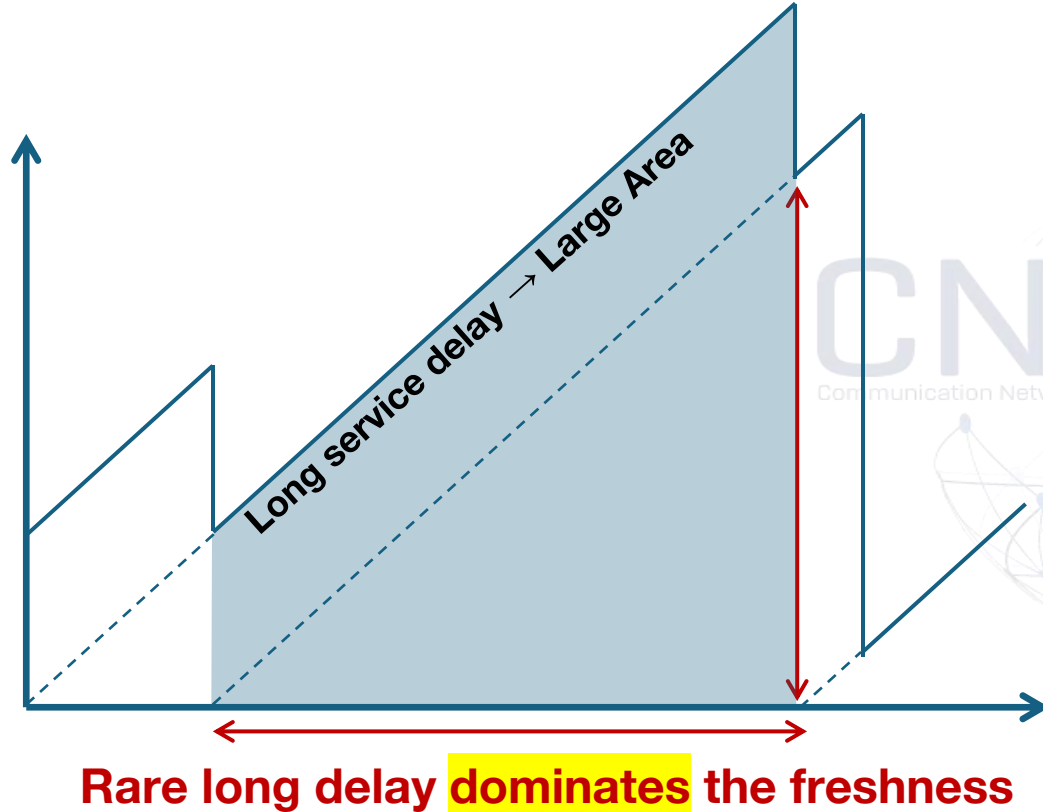


[1] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, "Update or Wait: How to Keep Your Data Fresh", *IEEE Transactions on Information Theory*, 2017.

[2] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, "Update or Wait: How to Keep Your Data Fresh", *IEEE INFOCOMM*, 2016. (**IEEE/ACM INFOCOM2026 Test of Time Paper Award**)

Taming the Heavy Tail: Age-Optimal Preemption

Challenges in Designing Age-Optimal Preemption

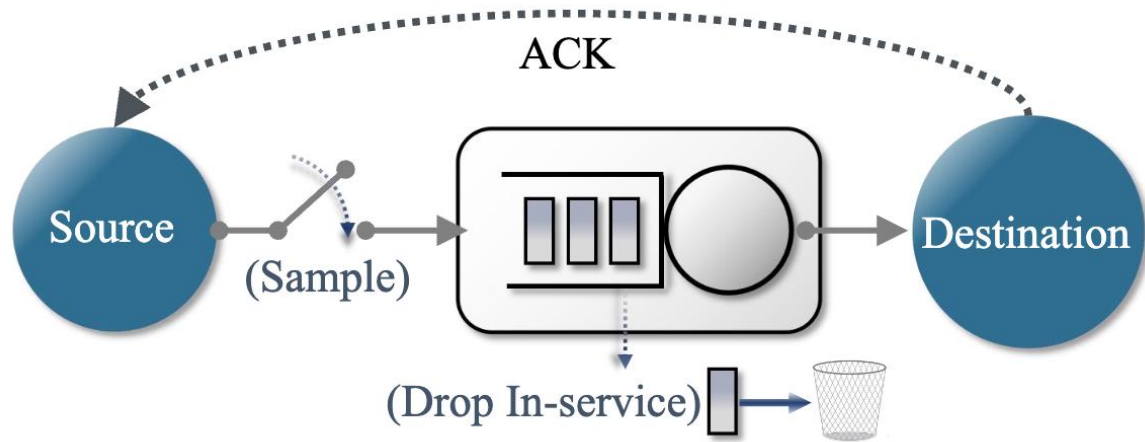


Challenges

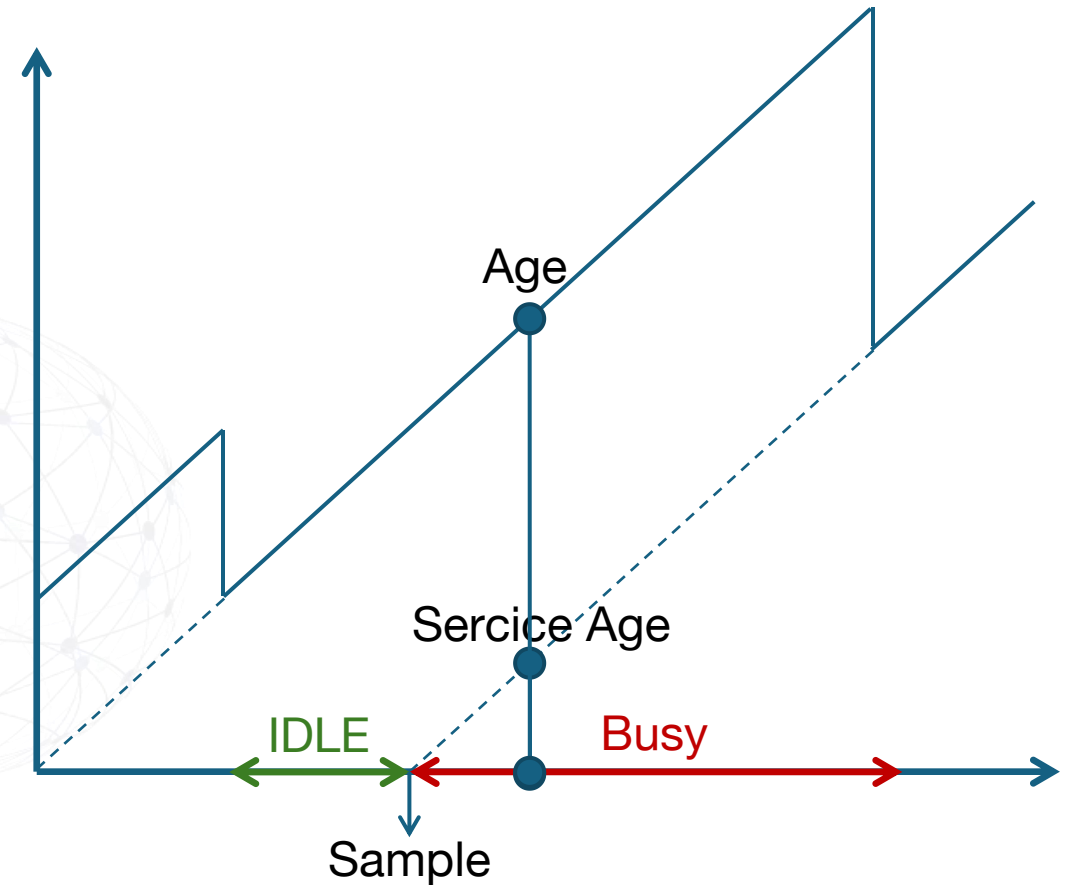
- Need to decide in continuous time.
- Decision epochs are *endogenous* and occur during service, not only at packet service completion.
- **MDP**: discrete decision epoch miss the continuous evolution of age.
- **SMDP**: random holding time help in designing age-optimal sampling, but do not naturally capture state-dependent interruption of the holding time.
- **We need a new Language**: Continuous-time Impulse Control Formulation

New Design Language for Preemption: **Impulse Control**

Hybrid State Space

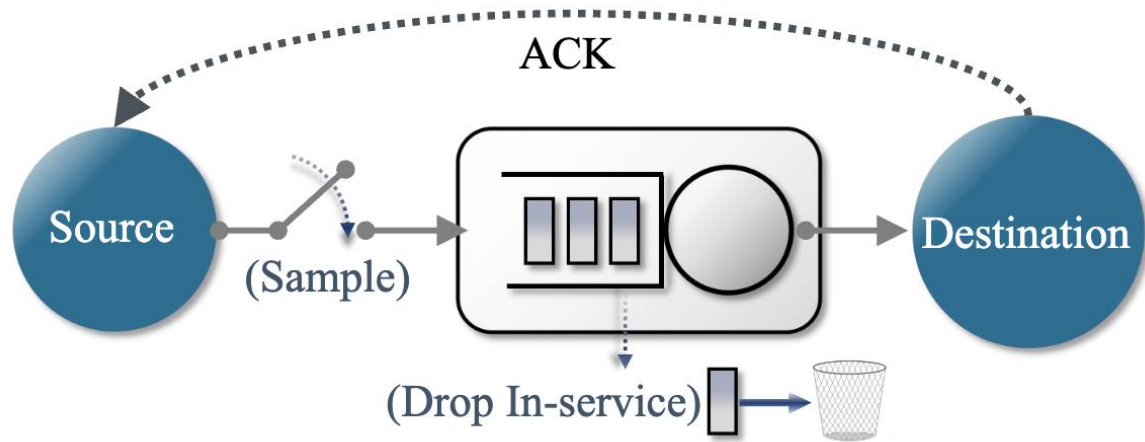


- Hybrid State Space: $(\Delta, b, m) = (\text{Age}, \text{Service Age}, \text{Mode})$
- **Idle mode:** $m = I$, the channel is idle and the source can decide whether to sample.
- **Busy mode:** $m = B$, the channel is busy and the source can decide whether to preempt and restart.

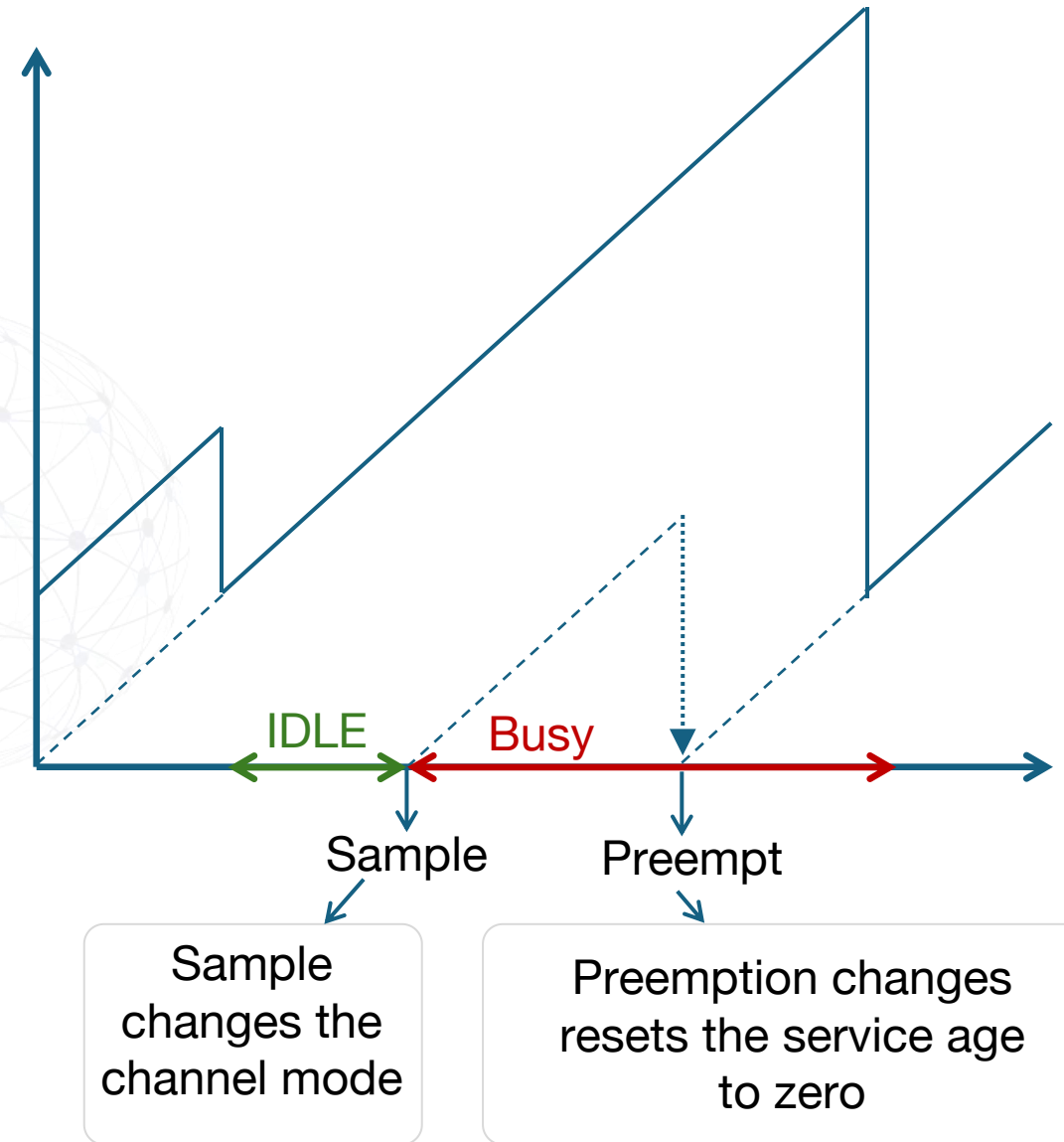


Design Language for Preemption: **Impulse Control**

Impulse Actions and Dynamics

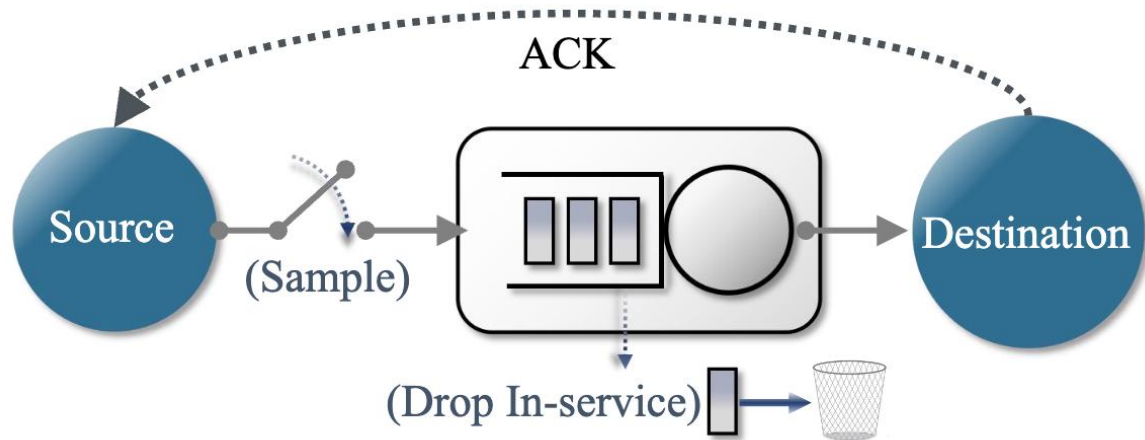


- **Sample:** It is available when channel is IDLE; It turns the channel from IDLE to BUSY:
 $(\Delta, 0, \text{IDLE}) \rightarrow (\Delta, 0, \text{BUSY})$
- **Preempt:** It is available when channel is BUSY; the channel is busy and the source can decide whether to preempt and restart.
 $(\Delta, b, \text{BUSY}) \rightarrow (\Delta, 0, \text{BUSY})$

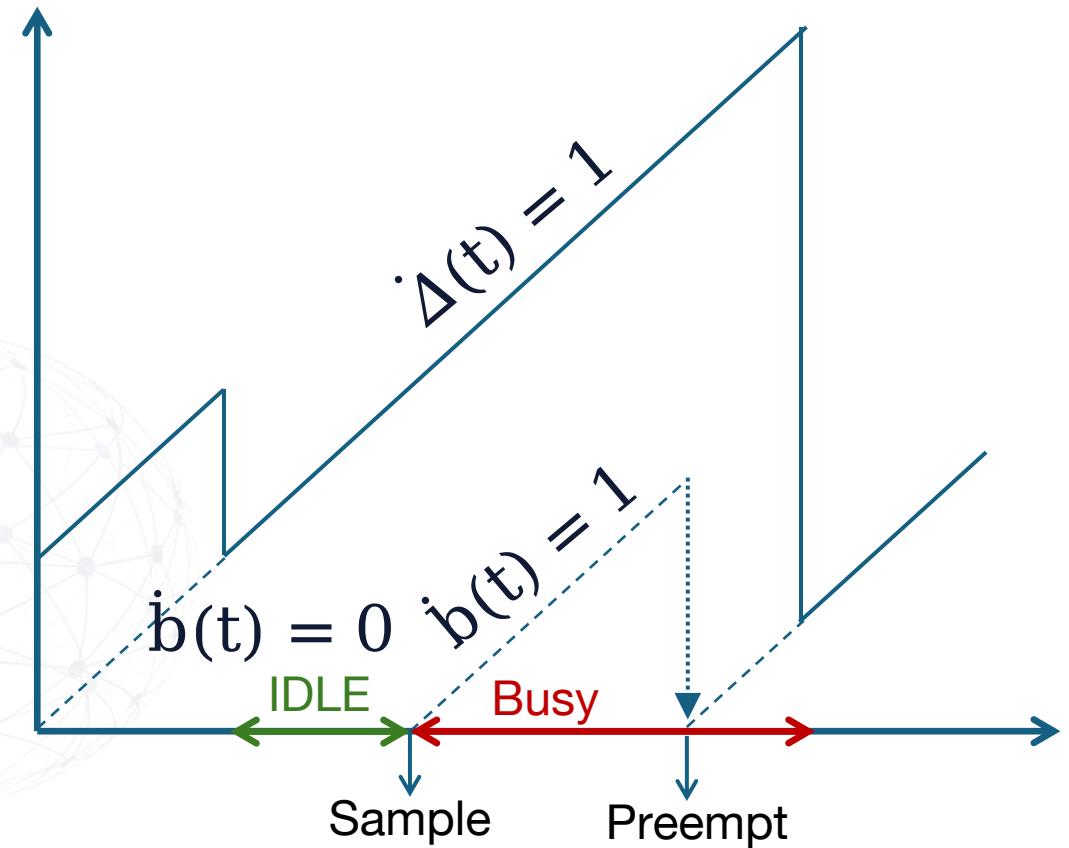


Method: Impulse Control Formulation

Deterministic Flows Between Impulse



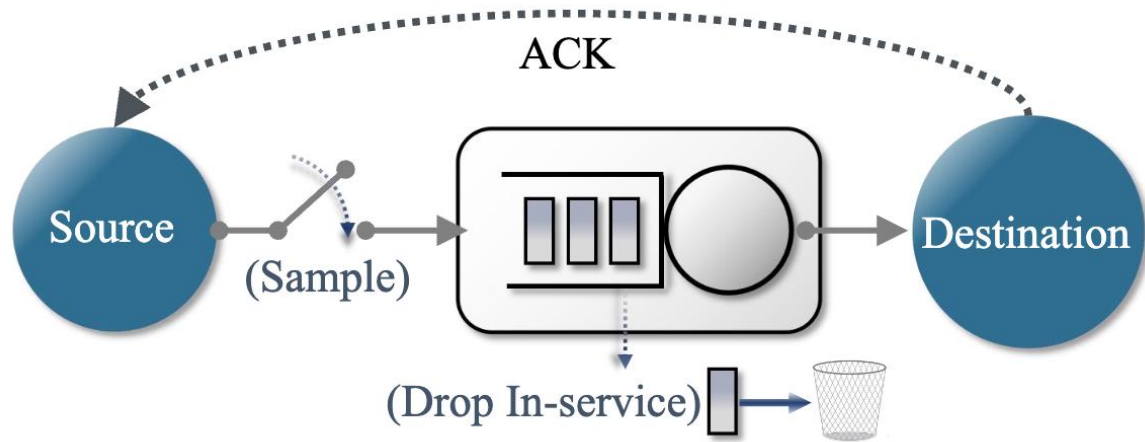
- Age: $\dot{\Delta}(t) = 1$
- Service Age: $\dot{b}(t) = \mathbb{I}_{\{m(t)=\text{BUSY}\}}$
- Running Cost:
 $g(\Delta, b, m) = \Delta$
- Sampling Cost: K_S
- Preemption Cost: K_P



If a service is completed, then the current age jumps from Δ to b

Method: Impulse Control Formulation

Objective Function



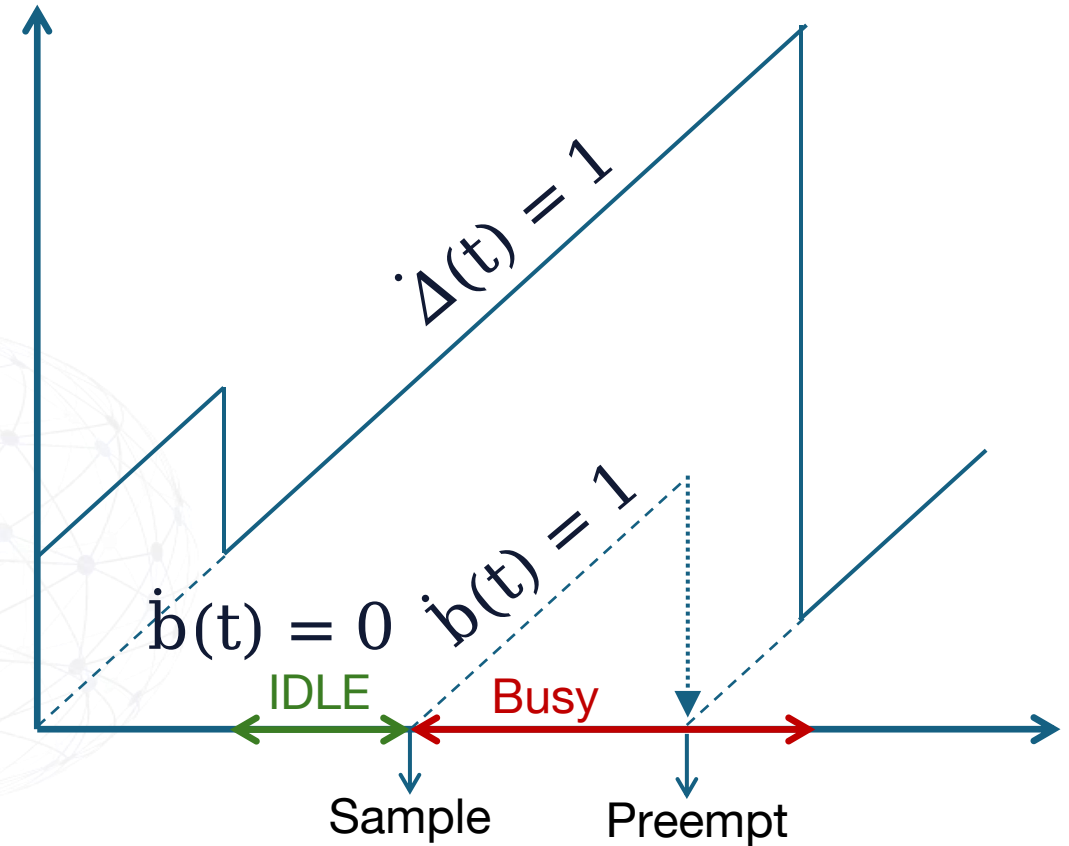
Minimize age plus sampling and preemption cost

$$\rho = \min_{\pi} \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_{\pi} \left[\int_0^T \Delta(t) dt + \sum_{i: \tau_i \leq T} K(X(\tau_i^-), a_i) \right]$$

Policy: when to sample and preempt

Average Age

Sampling and Preemption Cost



If a service is completed, then the current age jumps from Δ to b

Optimality Equations: Coupled Integral ACOEs (Theorem 1)

Objective Function

$$\rho = \min_{\pi} \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}_{\pi} \left[\int_0^T \Delta(t) dt + \sum_{i: \tau_i \leq T} K(X(\tau_i^-), a_i) \right]$$

Policy: when to sample and preempt

Average Age

Sampling and Preemption Cost

Coupled Integral ACOEs

- Optimal Sampling

$$z^*(\Delta) \in \arg \min_{z \geq \Delta} \left\{ \kappa_s + v(z) + \frac{z^2}{2} - \rho z \right\}$$

- Optimal Preemption

$$\theta^*(y) \in \arg \min_{\theta \geq 0} \left\{ Q(y, \theta; \rho) \right\}.$$

Busy Mode

For all $y = \Delta - b \geq 0$

$$v(y) = \inf_{\theta \geq 0} Q(y, \theta; \rho)$$

where $Q(y, \theta; \rho)$ is equal to:

$$Q(y, \theta; \rho) = (y - \rho)A(\theta) + J_1(\theta) + \int_0^\theta h_I(t) dF_Y(t) + \bar{F}_Y(\theta)(\kappa_P + v(y + \theta))$$

with $A(\theta) = \mathbb{E}[\min\{Y, \theta\}] = \int_0^\theta \bar{F}_Y(t) dt$ and $J_1(\theta) = \mathbb{E} \left[\int_0^{\min\{Y, \theta\}} t dt \right] = \int_0^\theta t \bar{F}_Y(t) dt$

IDLE Mode

For all $\Delta \geq 0$

$$h_I(\Delta) = \rho \Delta - \frac{\Delta^2}{2} + \kappa_P + \inf_{z \geq \Delta} \left\{ v(z) + \frac{z^2}{2} - \rho z \right\}$$

Optimality Equations: Coupled Integral ACOEs (Theorem 1)

Optimal Policy

- Optimal **Sampling**

$$z^*(\Delta) \in \arg \min_{z \geq \Delta} \left\{ \kappa_s + v(z) + \frac{z^2}{2} - \rho z \right\}$$

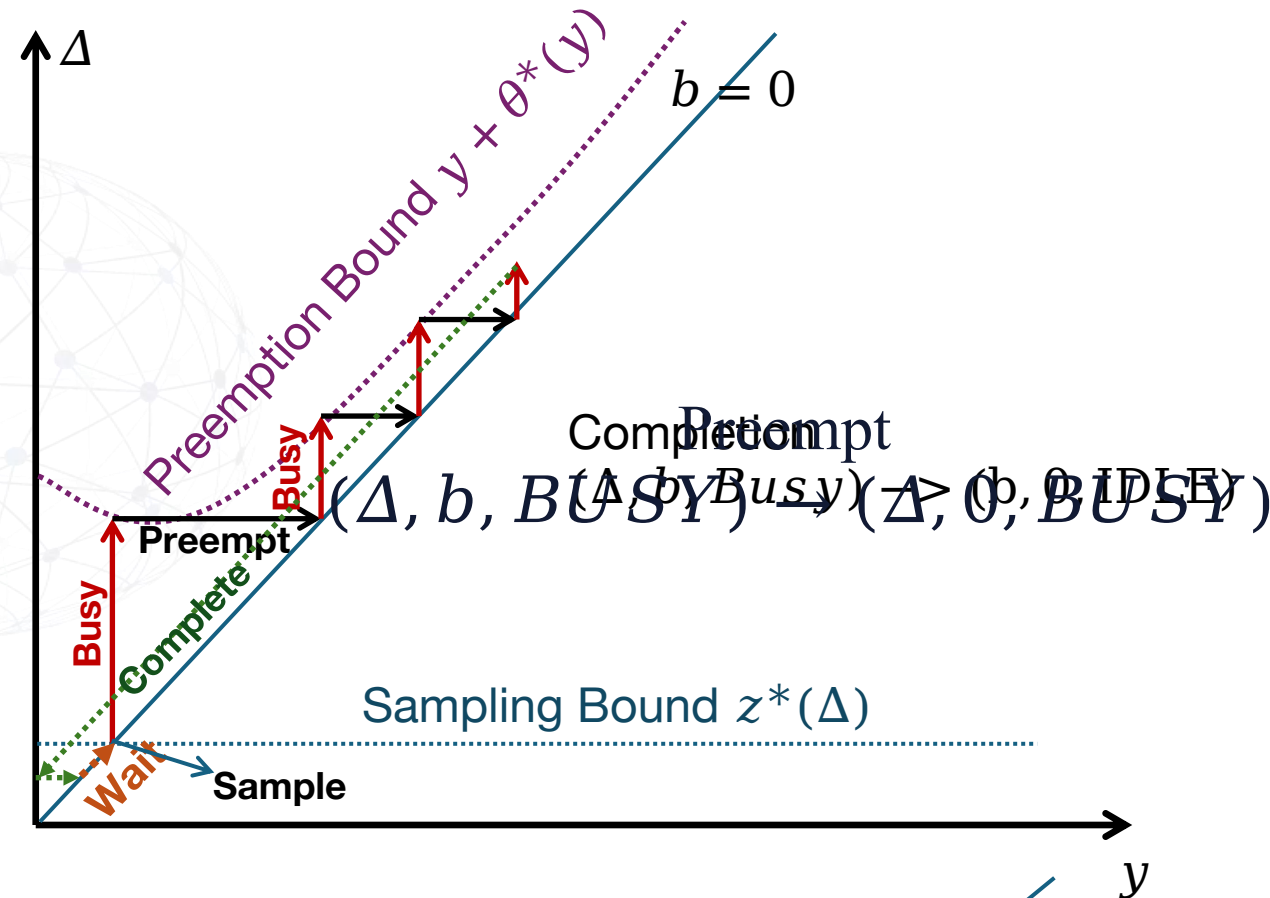
- Optimal **Preemption**

$$\theta^*(y) \in \arg \min_{\theta \geq 0} \left\{ Q(y, \theta; \rho) \right\}.$$

- IDLE: Wait until Age Δ hits $z^*(\Delta)$, then sample

- BUSY: Continue until completion or until service age b hits $\theta^*(y)$

System Dynamics



Difference between age and service age $y = \Delta - b$

Special Case Without Preemption (Corollary 1)

Busy Mode

For all $y = \Delta - b \geq 0$

$$v(y) = \inf_{\theta \geq 0} Q(y, \theta; \rho)$$

where $Q(y, \theta; \rho)$ is equal to:

$$Q(y, \theta; \rho) = (y - \rho)A(\theta) + J_1(\theta) + \int_0^\theta h_I(t) dF_Y(t) + \bar{F}_Y(\theta)(\kappa_P + v(y + \theta))$$

with $A(\theta) = \mathbb{E}[\min\{Y, \theta\}] = \int_0^\theta \bar{F}_Y(t) dt$ and
 $J_1(\theta) = \mathbb{E}\left[\int_0^{\min\{Y, \theta\}} t dt\right] = \int_0^\theta t \bar{F}_Y(t) dt$

Our
Result

Special Case of our Theory

Without preemption

- 1 If there is no preemption, i.e., $\theta = \infty$
- 2 **Surprise 1! Closed-form value function!**

$$v(y) = y\mathbb{E}[Y]$$

- 3 Optimal sampling policy:

$$z^*(\Delta) = \max\{\Delta, \rho - \mathbb{E}[Y]\}$$

Recovers the Update-or-Wait structure in Sun et al. [Theorem 4, TIT 2017].

IDLE Mode

For all $\Delta \geq 0$

$$h_I(\Delta) = \rho\Delta - \frac{\Delta^2}{2} + \kappa_S + \inf_{z \geq \Delta} \left\{ v(z) + \frac{z^2}{2} - \rho z \right\}$$

[1] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, "Update or Wait: How to Keep Your Data Fresh", **IEEE Transactions on Information Theory**, 2017.

Optimal Policy under Exponentially Distributed Delay (Theo. 2)

Busy Mode

For all $y = \Delta - b \geq 0$

$$v(y) = \inf_{\theta \geq 0} Q(y, \theta; \rho)$$

where $Q(y, \theta; \rho)$ is equal to:

$$Q(y, \theta; \rho) = (y - \rho)A(\theta) + J_1(\theta) + \int_0^\theta h_I(t) dF_Y(t) + \bar{F}_Y(\theta)(\kappa_P + v(y + \theta))$$

with $A(\theta) = \mathbb{E}[\min\{Y, \theta\}] = \int_0^\theta \bar{F}_Y(t) dt$ and
 $J_1(\theta) = \mathbb{E}\left[\int_0^{\min\{Y, \theta\}} t dt\right] = \int_0^\theta t \bar{F}_Y(t) dt$

IDLE Mode

For all $\Delta \geq 0$

$$h_I(\Delta) = \rho\Delta - \frac{\Delta^2}{2} + \kappa_P + \inf_{z \geq \Delta} \left\{ v(z) + \frac{z^2}{2} - \rho z \right\}$$

Our
Result

Special Case of our Theory

Exponential Delay

- 1 If $Y \sim \text{Exp}(\lambda)$.
- 2 **Surprise 2! Closed-form value function!**
- 3 Preemption policy: (Constant Threshold)

$$v(y) = y/\lambda$$

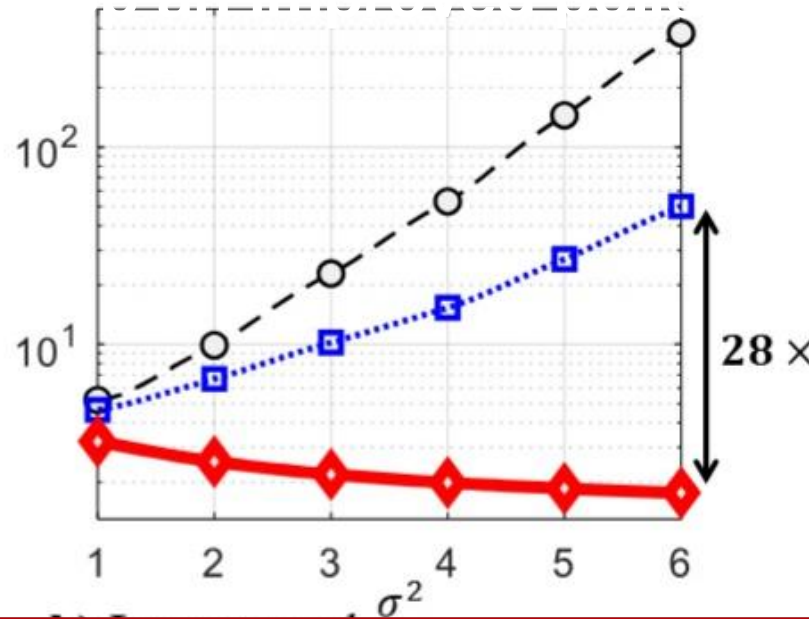
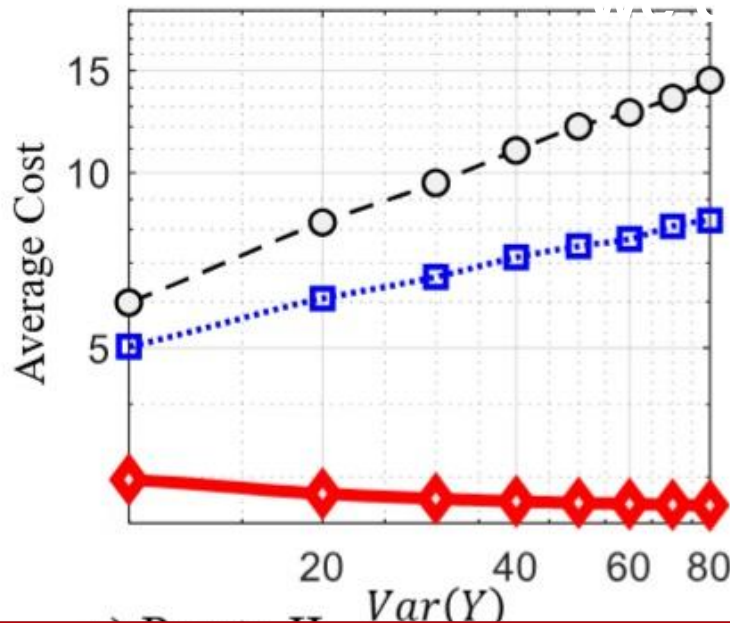
$$\theta^*(y) = c$$

Under Exponential Delay, we establish that the heuristic of [Allerton 2019] is optimal!

[Allerton 19] A. Arafa, R. D. Yates, and H. V. Poor, "Timely cloud computing: Preemption and waiting," in Proc. Annu. Allerton Conf. Commun. Control Comput. (Allerton), 2019, pp. 528–535.

Numerical Results and Insights

For general distributions, solving $v(y)$ in closed-form is challenging, we design policy iteration to numerically solve it.



- **28× Gains**

Compared to *Update or Wait* paper, our proposed policy reduces average cost by up to **28×**

- **Higher Variance → Lower Aol Cost**

As the variance of the delay increases, the Average cost decreases

With Preemption, Delay Variance can Become an Advantage of Information Freshness

[1] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, *IEEE Transactions on Information Theory*, 2017.

[2] Yin Sun, Elif Uysal, Roy D. Yates, C. Emre Koksal, Ness B. Shroff, “Update or Wait: How to Keep Your Data Fresh”, *IEEE INFOCOMM*, 2016. (**IEEE/ACM INFOCOM2026 Test of Time Paper Award**)

Taming the Heavy Tail: Age-Optimal Preemption

Takeaways



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Aimin Li · Yigit Ince · Elif Uysal

1 Heavy-tailed service delays make non-preemptive freshness control fragile.



2 Impulse control reveals a clean structure: busy-phase invariance and optimal stopping.



3 TAILOR delivers major gains and can even turn delay variance from a liability into an advantage.



Let introduce preemption to tame the heavy tail :)



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Thank you!
Questions?